

Hybrid Propulsion Systems for Small Ships: Context and Challenges

N. Bennabi, J. F. Charpentier, H. Menana, J. Y. Billard, P. Genet

Abstract -- New generations of ship must meet new challenges, particularly in terms of energy efficiency, reliability and environmental impacts. One of the future goals of shipbuilding is to reduce the impact of ship emissions to respond to existent and future regulations of the International Maritime Organization (IMO) on greenhouse gas and pollutants emissions. In this context, Hybrid Propulsion Systems (HPS)—are promising solutions to meet these requirements.

This work presents a state of the art of hybrid propulsion systems for small vessels. The paper includes a general description of the future challenges for shipbuilding and a description of the possible topologies of HPS which can be implemented in small ships. The key features of the HPS components are also described and several electric and hybrid propulsion small ship are presented

Index Terms—battery; CO₂ emissions; electrical power; fuel cell; hybrid Electric ship; parallel hybrid; series hybrid; supercapacitor.

I. NOMENCLATURE

IMO	International Maritime Organization
MARPOL	Marine Pollution
ECA	Emission Controlled Areas Sulfur

II. INTRODUCTION

THE transport sector consumes about 25% of total commercial energy consumed in the world [1].

Reducing energy consumption and emissions of transport systems are major issues because transportation sector have a strong impact on pollution, and fuel consumption. More than 80% of freight are transported by sea, and maritime transportation is responsible than more than 30% of the CO₂ emissions of the transportation sector and about 3% to 4% of the humanity CO₂ emissions.

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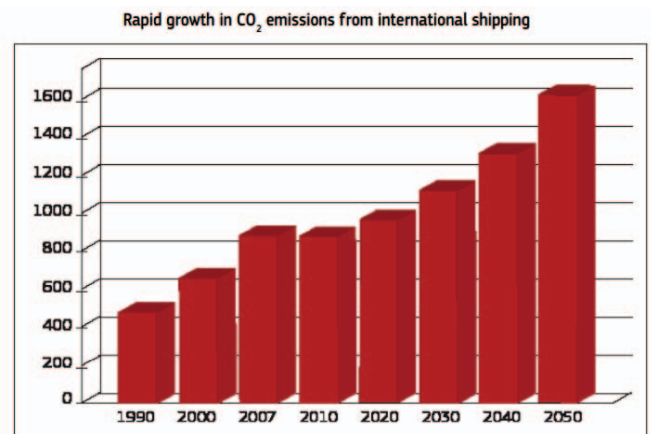
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Greenhouse gas emissions related to ships are important and rapidly growing. Without action, these emissions will be more than double by 2050, due to the expected growth of the global economy and the associated transport demand as shown in Fig. 1 [2]. This is why stringent international regulations have been established or are in project concerning limitation of ship emissions.



Estimated CO₂ emissions in million tonnes per year (data sources: 2nd IMO GHG study 2009 for emissions until 2007; MEPC 63/INF 2 for average scenario projections 2010-2050)

Fig. 1. The evolution of CO₂ emissions in shipping [2]

In this context, minimization of fuel consumption and reduction of emissions is one of the main objectives for designing new generations of ships [3]. One of the main challenges is to improve efficiency and optimally manage the energy/propulsion chain in order to reduce fuel consumption and environmental impact with an investment as low as possible [4]. In this perspective, electrification and hybridization of the propulsion chains are undeniably the solutions for the development of more efficient ships and environmental friendly ships [5]-[6].

In this paper, we will focus on the various hybrid configurations and the choice of energy generation and storage systems for small ships. Indeed, till today, due to technical and economic reasons, little interest was granted to electrification and hybridization of small ships.

III. REGULATORY ASPECTS

One of the future goals of the sea and river naval industry is to reduce the impact of boat emissions to best meet the existent and future stringent regulations of the International Maritime Organization (IMO) on emissions of greenhouse gases and pollutants.

The main emissions of ships are ozone particles, NO_x, sulfur dioxide, SO₂, and CO₂. These emissions are all oil

combustion products which can be classified as primary or secondary pollutants [7]. The primary pollutants are pollutants formed directly during the combustion process (sulfur oxide, oxide of carbon, nitrogen oxide...) while secondary pollutants are formed results of chemical reactions in the atmosphere (nitrogen dioxide, ozone, secondary particles...) [8].

MARPOL 73/78 adopted by the IMO, aims to prevent and mitigate at global marine pollution [9]. As an example, the annex VI of MARPOL defines strict limitations and emission controlled areas for Sulphur (ECAs) covering the English Channel, the North Sea, the North American coast and the Caribbean under US control area as shown in Fig. 2.



Fig.2. Emission Controlled Areas for Sulphur [10]

In these protected areas (ECAs) the maximum permissible Sulphur content of marine fuels is currently 1% and will fall to 0.1% in 2015. In other areas these Sulphur contents are subject to a limit of 3.5%. This overall limit will probably fall to 0.5% in 2025 while awaiting the results of a review to be completed in 2018 as shown in Fig. 3.

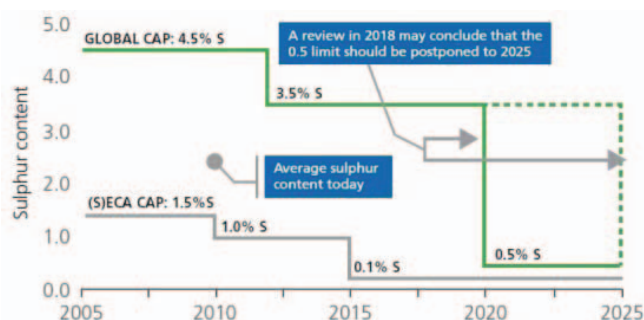


Fig.3. Sulphur content limits through time [11]

IV. HYBRIDIZATION OF SHIP PROPULSION AND REQUIREMENTS FOR SMALL SHIPS

New electric and hybrid propulsion architectures, allowed by innovative and high performance components (batteries, fuel cells, supercapacitors...) are a very promising track to reduce strongly pollutants and greenhouse gas emissions. If some studies are conducted for the integration of such architectures in large vessels, small units benefit only marginally today of this kind of systems. However, the integration of these new propulsion architectures on small ships operating on shorts cycles is a real technical challenge.

Indeed such vessel specifications imply to maximize their boarding capacity for passengers or freight to minimize the volume devoted to energy storage and energy/propulsion systems. Furthermore typical mission profiles, for this kind of vessel, include high transients of speed involving strong power demand. The available times for charging energy systems are often also limited. The aim of the following parts is to describe hybrid energy/propulsion topologies and to give some elements related to the adequacy of these topologies to these specific requirements.

A. General Advantages of electric and hybrid propulsion in the small ship context

Hybrid Propulsion Systems (HPS) are interesting when there is a large variation in power demand during operations. The design rules of this type of systems can differ significantly for different types of vessel, mainly due to the different types of mission profiles and operation requirements.

With all the currently available options, hybrid propulsion systems seem to be a very promising technologies to design high efficiency and environmental friendly small ships [12]-[13].

- HPS provide a better energy efficiency and significant greenhouse gas and pollutants emissions reduction than classical mechanical systems [14]-[11],
- Hybrid systems are modular, and offer a power redundancy.
- The use of electrical propulsion systems (POD system and thrusters) allow to improve maneuverability with important reduction of noise and vibrations [14],
- Electrical propulsion systems offers more degree of freedom in term of naval architecture and allows to free onboard available space.
- HPS allows to reduce of the level of vibration transmitted to the structure of the ship.
- HPS leads a reduction of maintenance of requirements on the combustion engines because the combustion engines are always optimally used.

However, all technologies are not mature and compromise cost, mass, and energy management performance is not yet satisfying. The following part describes the possible HPS topologies and underlines the significant pros and cons in the small ship context.

B. Hybrid propulsion topologies

Hybridization of the propulsion chains are based on the separation or simultaneous use of several different energy sources [7]. The choice of different sources, and sources combinations is designed to adapt to the ship power and energy requirements which include the needs of low and high power, the needs of flexibility in changes of rhythm and the needs of redundancy and reliability.

Use of hybrid system is a promising approach to minimize fuel consumption. Especially for vessels that requires a high degree of maneuverability and a very varied operating cycles [12]-[13]-[14]-[15].

There are three main architectures for hybrid propulsion chains.

1) Series Hybrid architecture

In this type of HPS architecture (Fig. 4), two kind of energy system (combustion engines and electric systems) are used in series [15]. The propulsion is done by a variable speed electrical drive (propulsion motor and power converter). The electrical power of the propulsion motor is generally supplied by a set of generators driven by combustion engines. Another option is to use fuel cells to provide electricity or a combination of fuel cells and generators driven by combustion engine. Energy storage systems (ESS) (battery, supercapacitors, etc.) can be connected to the electrical bus to provide propulsion power. This ESS can be used separately or with the generators [16].

In this configuration, electricity is used as an energy vector between the combustion engine and the propulsion motor, so there will be no direct mechanical connection between the engine and the propeller [17]. This configuration allows dissociating the operating points of the combustion engine and of the propeller. This is why it allows increasing the global efficiency of the system by optimizing the operating point of combustion engine and propeller in efficiency point of view. Of course, this advantage would be more significant if several generators and ESS are used, because it allows providing the required propulsion power by choosing an optimal combination of sources in order to keep the running combustion engines in their optimal operating area. However, for small ships, the possibility of multiplying the sources is very limited by volume and mass constraints.

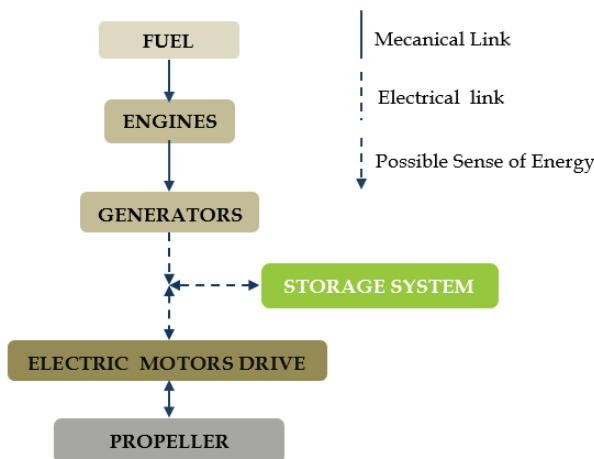


Fig.4. Typical Series hybrid architecture [18]

2) Parallel Hybrid architecture

A typical parallel hybrid propulsion schema is shown in Fig.5. Parallel hybrid architecture combines two kinds of propulsion motors: electric motors and combustion engines which are mechanically coupled by the same shafts through gearboxes and clutches. The electrical motors and drive are supplied by electrical ESS and/or independent power sources (generators driven by combustion engines and/or Fuel Cell

for example). The two propulsion systems can be used together or separately [19].

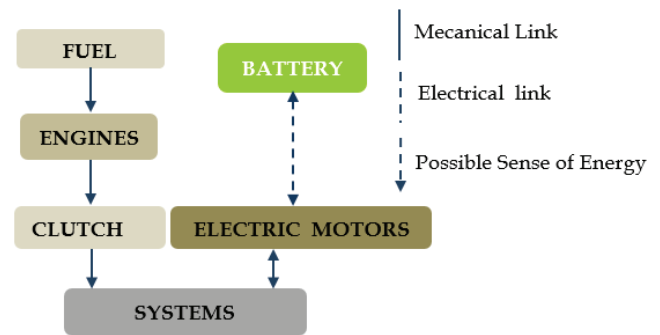


Fig.5. Typical Parallel hybrid architecture [18]

In most cases one of the propulsion systems is devoted to be used for low speed operations (classically the electric system) and the other one (classically the combustion engine) is used for high speeds. The first system can also be used as a boost system to provide a supplementary power during transient (starting, acceleration, etc.). This architecture reduces the number of elements compared to hybridization series and allows optimizing the sizing of each of the energy sources [20].

3) Series-Parallel Hybrid architecture

The series-parallel architecture combines both architectures defined previously. This association allows the passage of a parallel-type operation to a serial type operation [21]-[17].

Table I gives a comparison between the series and parallel hybridization architectures, in terms of costs, redundancy and efficiency:

TABLE I
PERFORMANCES OF HYBRID PROPULSION [15]-[21]-[17]-[19]

	Series	Parallel
Sizing of element s/costs	- To be fully efficient, the system must combine several power sources with rated power equal to a fraction of total power. It leads to supplementary costs volumes and weight. - Electric propulsion motors sizing corresponds to the peak power	- The system allows optimizing the sizing of each of the propulsion systems. - Electrical motor sizing is reduced. - Supplementary costs and volume related to complex mechanical system (clutches, gearboxes)

Power and Efficiency	- The efficiency is limited by the multiplication of systems in series (engine, generator, transformer, converters, motors, etc). - The system allows a precise control and gradual variation of the propeller speed. If several sources are used, combustion engines can be operated at peak efficiency	- Good efficiency related to the reduced numbers of series components (particularly in combustion engine mode) - Precise control and a gradual variation of the propeller speed. - Combustion engine operates at peak efficiency
Redundancy/reliability	The system is modular, and offers power sources redundancy	- Very high level of redundancy since the propulsion power can be provided independently by two independent systems. - Complex mechanical system which can be a cause of failure.

C. Power sources in HPS

HPS systems for ships are based on the combination of several power sources and energy storage systems. This part aims to presents the main power sources and energy storage systems which can be used in HSP. The minimization of the fuel consumption on operating cycle requires the knowledge of the behavior of these different energy sources.

1) Power generation systems

a) Diesel Engines

Diesel engines remain currently the most used engines for ships. This power source is characterized by a high level of autonomy and by a relatively low cost compared to other sources. Diesel engines are characterized by a high level of greenhouse gases and pollutants emission particularly if they are used at a lower level of power than their rated power. Several attempts have been made to reduce emissions of the diesel engines; one of the possible solutions is the use of Liquefied Natural Gaz, LNG.

In recent years, the use of LNG as an alternative fuel has been tested and studied [22]. Fig.6 shows a comparison of the use of several kinds of fuel and of LNG in terms of emissions.

These results shows that LNG is very interesting alternative to diesel for reducing emissions, with a reduction of about 85-90% of NOx emissions, almost 100% of SOx emissions and about 15-20% of CO₂ emissions.

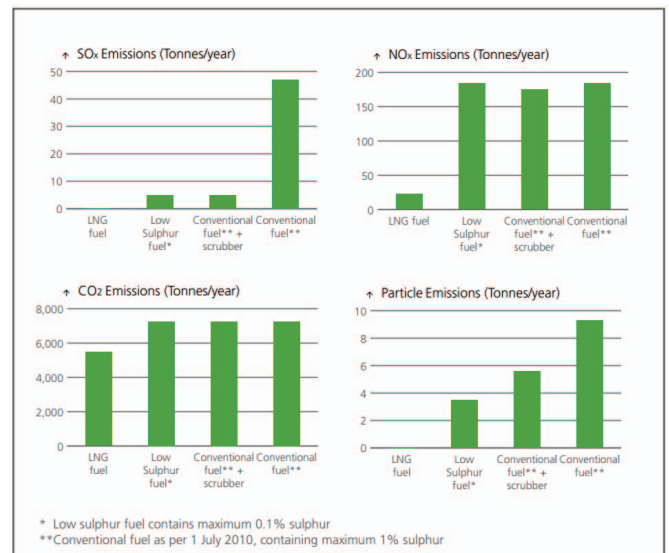


Fig.6. Emissions fuel for typical boats from [22]

b) Fuel cell

Fuel cells can be an interesting alternative for power production on small ships with very low emissions.

A fuel cell has the life expectancy of a diesel engine. It is an electrochemical device that converts chemical energy of a fuel into electrical energy [23]-[24]-[25].

Fuel cells are classified according to operating temperature and the nature of the used electrolyte. There are two categories of fuel cells: high-temperature and low-temperature fuel cell [24]-[26]. Low temperature fuel cell categories include the Alkaline Fuel Cells (AFC), the Proton Exchange Membrane Fuel Cells (PEMFC), the Direct Methanol Fuel Cells (DMFC) and the Phosphoric Acid Fuel Cells (PAFC). These Fuel Cells are characterized by an operating temperature from 50 to 250°C [17]-[25]-[28]. High temperature fuel cells include the Solid Oxid Fuel Cells (SOFC) and the Molten Carbonate Fuel Cells (MCFC). Their operating temperatures are comprised between 650 abd 1000°C [23]-[24]-[25].

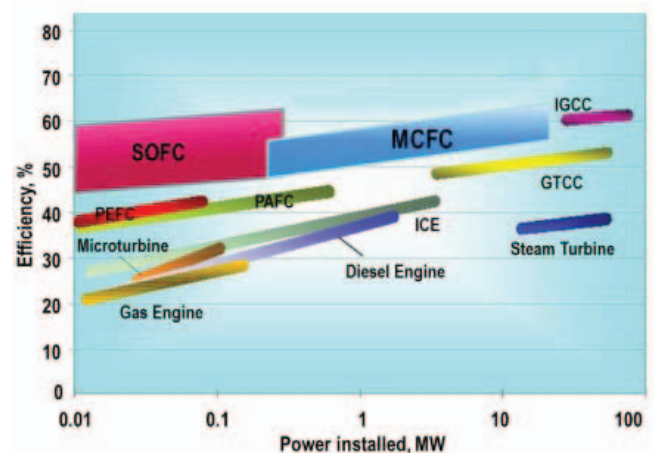


Fig.7. Comparison of the efficiency and electrical power installed for the combustion-based systems and fuel cell systems [29]

The Fig.7 shows a comparison between the different elements of generation of energy, and shows that PEMFC have an efficiency up to 45% (efficiency of PSCF is about 40% and diesel engine efficiency is about 35%). It can provide high efficiency power with zero emission.

It can be also noticed that PEMFC technology is characterized by relatively low operating temperature which ensures an easy and quick start what could be interesting for vessels with a variable duty cycle [17]-[30]. Another advantage is its significant lifetime compared to other fuel cells types. All this points demonstrates that PEMFC can be a very attractive element for HPS for low power small ship.

2) Electric storage system

An intermediate energy storage device can be used to better manage energy and reduce the effects of peak power level on the sizing of the energy sources. The integration of energy storage systems offers interesting solutions for the efficiency and optimal use of energy. It can also lead to a significant reduction in noise and vibration compared to conventional systems [30]-[31].

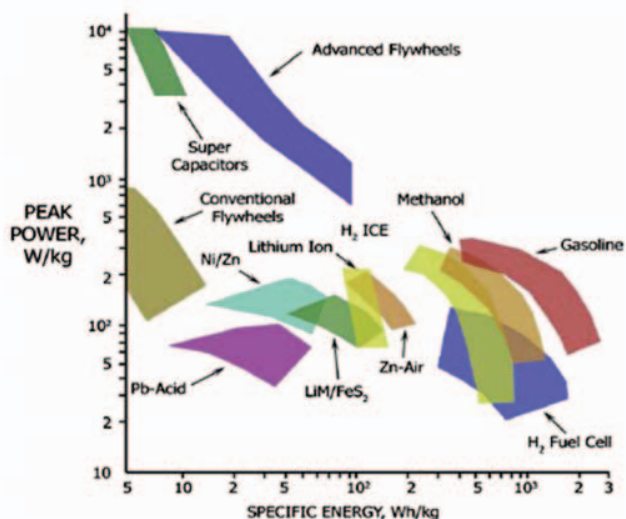


Fig.8. Ragone diagram of several storage systems [32]

Fig.8 gives a comparison of different energy storage systems in terms of specific energy and peak power.

The oldest and most classical method for storing electrical energy is using batteries. As shown in Fig. 8 lithium-ion batteries presents significant advantages compared to other battery types, particularly in term of energy density and efficiency [30].

Supercapacitors can also be interesting intermediate energy storage systems for small ships with short power transients. They are characterized by a high value of charge and discharge peak power and can be used to rapid dynamic storage with very low specific energy. They are usually associated with slower systems, such as batteries and fuel cells, to manage high power short transients [30]-[31]. They can operate at very low temperatures at which many types of electrochemical batteries stop working.

Flywheels can be also used in some particular cases. Like supercapacitors, they are characterized by a high power density but a relatively low energy density. However using

flywheels leads to implement safety systems to protect crew and passengers from possible flywheel failure. So their use in small ship context seems very limited.

D. Examples of electrical and hybrid vessels

1) Nemo H2 project

The project was launched in Amsterdam in 2009. It is one of the first boats developed using fuel cells. Nemo H2 is 22 meters in length and can carry 86 passengers. The propulsion is based on a series architecture where power is supplied by a fuel cell system PEMFC (60-70kW) [21]-[33].

When the ship is not at full power, batteries lead-acid (30-50kW) are charged. These batteries can then provide extra power when needed [33]-[34].



Fig.9. Nemo H2 Amsterdam [18]

2) Hybrid TUG RT Adrian

A new generation of hybrid propulsion tug has been built in 2011. This first European hybrid tug, called RT Adrian (Fig.10) is used in the port of Rotterdam. The tug hybrid architecture, called "XeroPoint Hybrid Propulsion System" is a series/parallel hybrid architecture combining diesel engines, battery pack and diesel generators as shown in Fig.11. The propulsion system can be used in all-electric or diesel electric mode (for low power modes), conventional mode in the mid power range or in combined mode for full power. In this combined mode, diesel propulsion engines, and electrical motors, are used simultaneously.



Fig.10. Hybrid TUG RT Adrian in the port of Rotterdam [8]

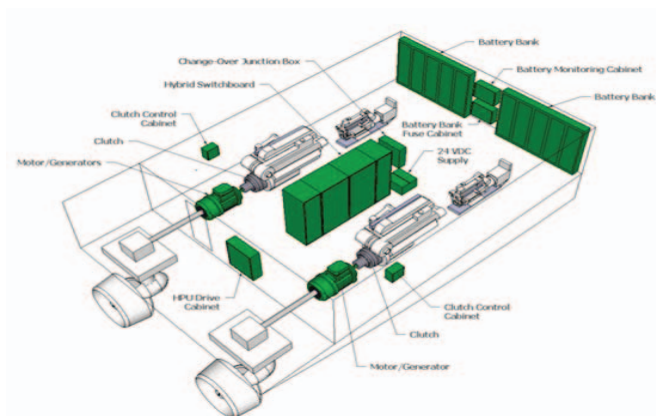


Fig.11. Xeropoint Schematic propulsion system (source AKA from [8])

3) *Bus electro-solar sea in La Rochelle*

This hybrid architecture boat can carry 75 passengers' at a speed of 6 knots. It is equipped with photovoltaic panels to assist the diesel engine to charge the batteries.



Fig.12. Bus electro-solar sea in La Rochelle [3]

4) *The ar Vag Tredan*

This boat of 22.1 meters in length and 1.7 meters in draft has been built by the STX France shipyard in Lorient. Ar Vag Tredan is powered by supercapacitors. These super capacitors are used to provide power to two azimuth thrusters of 70kW. The ship can cruise at 10 knots during 2x7mn and ensure zero emissions operations. At each stop the 8 sets of 16 supercapacitors are recharged in 4 minutes [35].



Fig.13. The Ar Vag Tredan [35]

5) *BatCub*

BatCub project consists in using two catamaran shape water buses of 19 meters in length. This particular hull shape allows to increase seakeeping. Energy/propulsion system is based on a series hybrid propulsion (diesel-electric), which mainly operates on the electric mode. Each shuttle is equipped with a 140kWh lithium ion battery system. The system powers two electric motors for propulsion and auxiliary systems such as lighting and communications. Coupled to diesel engine/generator, the battery is able to store the energy produced by the generator and provide additional propulsive power when needed. Battery can be combined to diesel/generators during the boost phases,

During night, when the boat is docked, the batteries are charged by the harbor power grid. When they are fully charged the batteries can provide 6 hours of autonomy in all-electric operations.



Fig.14. The BatClub [36]

V. CONCLUSION

This paper presents a state of the art of hybrid propulsion systems which can be used for small ship application. After a brief description of the main challenges for the design of the energy/propulsion systems for small ships, several hybrid architectures for small ship are presented, highlighting the key components. Several examples of small ships with hybrid and electric propulsion have also been presented.

This paper shows that Hybrid Propulsion Systems appears to be a very interesting solution to optimize the behavior of small and low power ship in terms of efficiency and greenhouse gas and pollutant emissions.

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VII. BIOGRAPHIES

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